EXP:- 6

ERROR DETECTION AND CORRECTION USING HAMMING CODE

AIM:-

To write a program to implement error detection and correction using the hamming code

ALGORITHM:

1. Start the program
2. Read the number of 4 bit data block (n) forming the input data stream
3. For each 4-bit data block D = d1 d2 d3 d4, Perform hamming encoding
   a) Arrange 7 bits with parity at positions, 1, 2 and 4 and data bits at position 3, 5, 6 and 7
   b) Compute the parity bits at position
      $P_1 = d_1$ XOR d_2 XOR d_4 (Cover bit 1, 3, 5,
      $P_2 = d_1$ XOR d_3 XOR d_4 (Cover bit 2, 3, 6
      $P_3 = d_2$ XOR d_3 XOR d_4 (Cover bit 4, 5,
4. Simulate transmission by introducing a single bit error at a chosen bit Position in the Code word.

```
=== HAMMING CODE - ERROR DETECTION AND CORRECTION ===

Enter number of 4-bit data blocks in the stream: 2

--- Block 1 ---
Enter 4 data bits (space separated, e.g. 1 0 1 1): 1 0 1 1
Original Data Bits : 1011
Encoded Hamming Code (7 bits): 0110011
Enter bit position to introduce error (1-7, 0 = no error): 5
Transmitted Code (with error induced at pos 5): 0110111
Received Code : 0110111
Checking for errors...
 Error detected at bit position: 5 -> Correcting...
Corrected Code : 0110011
Recovered Data Bits : 1011

--- Block 2 ---
Enter 4 data bits (space separated, e.g. 1 0 1 1): 1 0 1 1
Original Data Bits : 1011
Encoded Hamming Code (7 bits): 0110011
Enter bit position to introduce error (1-7, 0 = no error): 0
Transmitted Code (no error induced): 0110011
Received Code : 0110011
Checking for errors...
 No error detected.
Corrected Code : 0110011
Recovered Data Bits : 1011
```

5. At the receiver recompute the check bits from the recieved code word.

$C_1 = P_1$ XOR d_1 XOR d_2 XOR d_4

$C_2 = P_2$ XOR d_1 XOR d_3 XOR d_4

$C_4 = P_4$ XOR d_2 XOR d_3 XOR d_4

6. Compute Error Position = C_4 C_2 C_1

7. If error Position = 0, no error has Occurred

8. Extract the original 4 data is inverted to correct the error.

9. Repeat Steps 3 to 8 for all n blocks in the data stream

10. Stop the program

PROGRAM:

```
def encode(d):
    """d: list of 4 bit [d1,d2,d3,d4] -> returns 7 bit hamming code list Positions (1-indexed): 1=P1 2=P2 3=P3 4=P4 5=d2 6=d3 7=d4"""
    h = [0] * 7
    h[2] = d[0]
    h[4] = d[1]
    h[5] = d[2]
    h[6] = d[3]
```

RESULT:-

The program to implement error detection and correction using the hamming code concept was successfully written in Python, executed and verified using a sample input data stream.